EST II — MATH LEVEL 2 • UNIT 8.01

C Trigonometric Laws, Polar Coordinates & Conic Sections

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

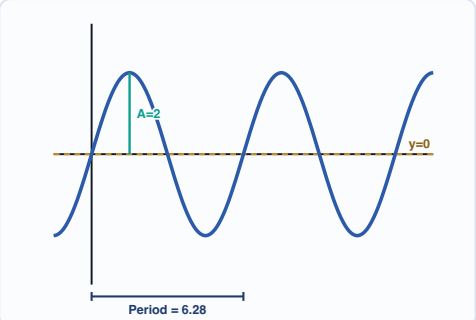
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 B	Q9 C	Q10 D
Q11 D	Q12 D	Q13 A	Q14 C	Q15 C
Q16 A	Q17 C	Q18 C	Q19 A	Q20 A
Q21 C	Q22 B	Q23 A	Q24 D	Q25 C

Trigonometric Laws, Polar Coordinates & Conic Sections Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?



A $2 + 3i$ ✓

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

WORKED SOLUTION & VERIFICATION

To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is $2 + i$:

$$\frac{7+4i}{2-i} = \frac{(7+4i)(2+i)}{(2-i)(2+i)}$$

Expand the numerator using the distributive property (FOIL):

$$(7+4i)(2+i) = 14 + 7i + 8i + 4i^2$$

Since $i^2 = -1$:

$$14 + 15i + 4(-1) = 14 - 4 + 15i = 10 + 15i$$

Expand the denominator using the difference of squares:

$$(2-i)(2+i) = 2^2 - i^2 = 4 - (-1) = 5$$

Now, divide each term in the numerator by the denominator:

$$\frac{10+15i}{5} = \frac{10}{5} + \frac{15i}{5} = 2 + 3i$$

Common Misconceptions:

- Choice B occurs from an arithmetic sign error in the imaginary term.
- Choice C occurs if the student mistakenly calculates the denominator as $2^2 - 1^2 = 3$ by forgetting that $-i^2 = -(-1) = +1$.
- Choice D occurs if the student takes $i^2 = 1$, yielding $14 + 4 = 18$ in the numerator.

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

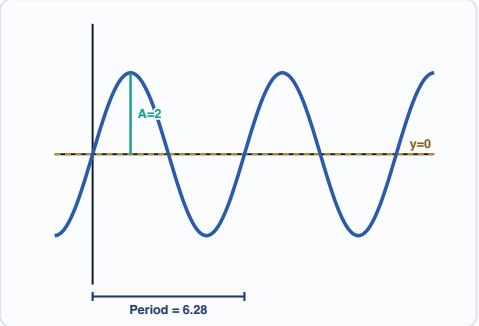
- **A** 0 ✓
- **B** 2
- **C** -1
- **D** 3

WORKED SOLUTION & VERIFICATION

The value of $f(1)$ corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point $(1, 0)$ is marked. Therefore, $f(1) = 0$.

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?



- **A** (1, 0) and (3, 0) ✓
- **B** (0, 1) and (0, 3)
- **C** (-1, 0) and (-3, 0)
- **D** (1, 0) and (-3, 0)

WORKED SOLUTION & VERIFICATION

The x-intercepts of a function occur where the function's value is zero, i.e., $f(x) = 0$. \nWe need to solve the quadratic equation $x^2 - 4x + 3 = 0$. \nThis equation can be factored as $(x - 1)(x - 3) = 0$. \nTherefore, the solutions are $x = 1$ and $x = 3$. \nThis means the x-intercepts are $(1, 0)$ and $(3, 0)$.

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

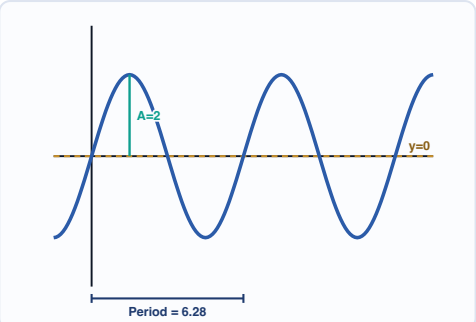
- **A** [3, 4] ✓
- **B** [5, 8]
- **C** [-3, -4]
- **D** [4, 3]

WORKED SOLUTION & VERIFICATION

The components of a vector $\mathbf{v}$ represented by a directed line segment from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$ are given by $\mathbf{v} = [x_2 - x_1, y_2 - y_1]$. In this case, $P = (1, 2)$ and $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $[4 - 1, 6 - 2] = [3, 4]$.

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?



- **A** $\frac{3}{5}$ ✓
- **B** $\frac{2}{5}$
- **C** $\frac{1}{2}$
- **D** $\frac{3}{4}$

WORKED SOLUTION & VERIFICATION

The total number of marbles in the bag is the sum of red and blue marbles: $12 + 8 = 20$ marbles. The number of favorable outcomes (drawing a red marble) is 12. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is $P(\text{red}) = \frac{\text{Number of red marbles}}{\text{Total number of marbles}} = \frac{12}{20}$. This fraction can be simplified by dividing both the numerator and the denominator by their greatest common divisor, which is 4: $\frac{12 \div 4}{20 \div 4} = \frac{3}{5}$. Thus, the probability of drawing a red marble is $\frac{3}{5}$.

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

- **A** 14 ✓
- **B** 17
- **C** 8
- **D** 11

WORKED SOLUTION & VERIFICATION

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Plugging these values into the formula, we get: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the 4th term of the sequence is 14.

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

- **A** 96 ✓
- **B** 192
- **C** 98
- **D** -96

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(2)^4 = 96$. Therefore, the verified correct option is 96.

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

B -2 ✓

C 2

D 1

WORKED SOLUTION & VERIFICATION

The general form of the quadratic function is $y = ax^2 + bx + c$. We are given three points that lie on the graph: $(-1, 5)$, $(1, 1)$, and $(3, 5)$.

Substituting these points into the equation, we get a system of three linear equations with three unknowns (a, b, and c):

1. For $(-1, 5)$: $a(-1)^2 + b(-1) + c = 5 \Rightarrow a - b + c = 5$
2. For $(1, 1)$: $a(1)^2 + b(1) + c = 1 \Rightarrow a + b + c = 1$
3. For $(3, 5)$: $a(3)^2 + b(3) + c = 5 \Rightarrow 9a + 3b + c = 5$

We can solve this system of equations. Let's subtract equation (1) from equation (2):

$(a + b + c) - (a - b + c) = 1 - 5$

$2b = -4$

$b = -2$

Alternatively, notice that the y-values for $x = -1$ and $x = 3$ are the same (both are 5). This indicates that the axis of symmetry of the parabola is exactly in the middle of $x = -1$ and $x = 3$. The x-coordinate of the vertex (axis of symmetry) is given by $-\frac{b}{2a}$.

Axis of symmetry: $x = \frac{-1+3}{2} = \frac{2}{2} = 1$

So, $-\frac{b}{2a} = 1$, which implies $b = -2a$.

Now, substitute this into equation (2): $a + b + c = 1$

$a + (-2a) + c = 1$

$-a + c = 1 \Rightarrow c = a + 1$

Substitute $b = -2a$ and $c = a + 1$ into equation (1):

$a - (-2a) + (a + 1) = 5$

$a + 2a + a + 1 = 5$

$4a + 1 = 5$

$4a = 4$

$a = 1$

Now we can find b using $b = -2a$:

$b = -2(1)$

$b = -2$

To confirm, we can also find c: $c = a + 1 = 1 + 1 = 2$. The function is $y = x^2 - 2x + 2$.

Let's check the points:

For $(-1, 5)$: $(-1)^2 - 2(-1) + 2 = 1 + 2 + 2 = 5$ (Correct)

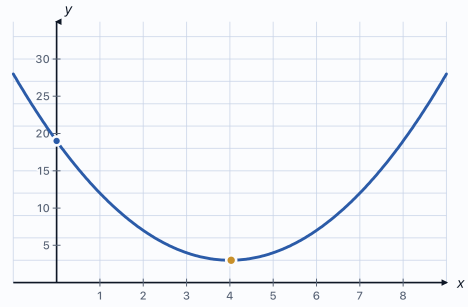
For $(1, 1)$: $(1)^2 - 2(1) + 2 = 1 - 2 + 2 = 1$ (Correct)

For $(3, 5)$: $(3)^2 - 2(3) + 2 = 9 - 6 + 2 = 5$ (Correct)

The value of b is -2.

9 MCQ

In Trigonometry, Polar & Conic Sections (problem set #925), consider the function $f_{925}(x) = x^2 - 8x + 19$ illustrated on the coordinate axes. Express $f_{925}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{925}(6)$.



A 14

B 7

C 10 ✓

D 8

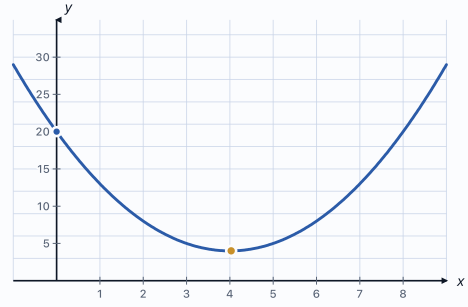
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{925}(x) = (x-4)^2 + 3$, so $k = 3$. **[M1]**
2. Evaluate $f_{925}(6) = (6-4)^2 + 3 = 4 + 3 = 7$. **[M1]**
3. Sum: $k + f_{925}(6) = 3 + 7 = 10$. **[A1]**

10 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #926), consider the function $f_{926}(x) = x^2 - 8x + 20$ illustrated on the coordinate axes. Express $f_{926}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{926}(7)$.



A 21

B 13

C 14

D 17 ✓

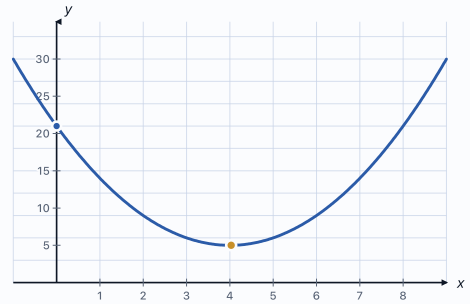
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{926}(x) = (x-4)^2 + 4$, so $k = 4$. **[M1]**
2. Evaluate $f_{926}(7) = (7-4)^2 + 4 = 9 + 4 = 13$. **[M1]**
3. Sum: $k + f_{926}(7) = 4 + 13 = 17$. **[A1]**

11 MCQ

Examining Trigonometry, Polar & Conic Sections (problem set #927), consider the function $f_{927}(x) = x^2 - 8x + 21$ illustrated on the coordinate axes. Express $f_{927}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{927}(8)$.



A 21

B 22

C 30

D 26 ✓

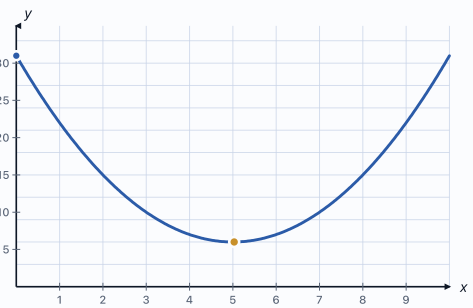
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{927}(x) = (x-4)^2 + 5$, so $k = 5$. **[M1]**
2. Evaluate $f_{927}(8) = (8-4)^2 + 5 = 16 + 5 = 21$. **[M1]**
3. Sum: $k + f_{927}(8) = 5 + 21 = 26$. **[A1]**

12 MCQ

For the configuration in Trigonometry, Polar & Conic Sections (problem set #928), consider the function $f_{928}(x) = x^2 - 10x + 31$ illustrated on the coordinate axes. Express $f_{928}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{928}(10)$.



A 32

B 42

C 31

D 37 ✓

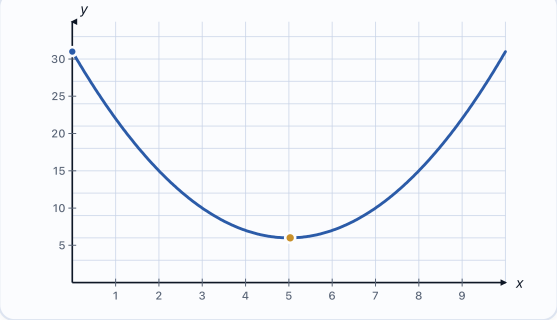
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{928}(x) = (x-5)^2 + 6$, so $k = 6$. **[M1]**
2. Evaluate $f_{928}(10) = (10-5)^2 + 6 = 25 + 6 = 31$. **[M1]**
3. Sum: $k + f_{928}(10) = 6 + 31 = 37$. **[A1]**

13 MCQ

Analyzing Trigonometry, Polar & Conic Sections (problem set #929), consider the function $f_{929}(x) = x^2 - 10x + 32$ illustrated on the coordinate axes. Express $f_{929}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{929}(11)$.



- **A** 50 ✓
- **B** 44
- **C** 55
- **D** 43

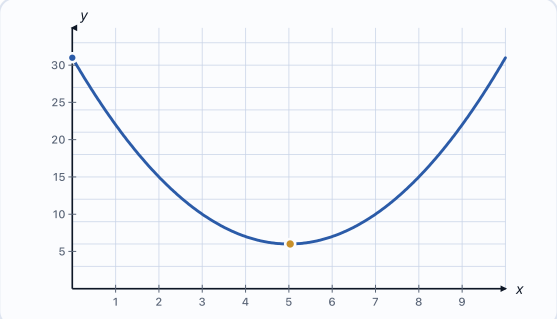
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{929}(x) = (x-5)^2 + 7$, so $k = 7$. **[M1]**
2. Evaluate $f_{929}(11) = (11-5)^2 + 7 = 36 + 7 = 43$. **[M1]**
3. Sum: $k + f_{929}(11) = 7 + 43 = 50$. **[A1]**

14 MCQ

Within Trigonometry, Polar & Conic Sections (problem set #930), consider the function $f_{930}(x) = x^2 - 10x + 33$ illustrated on the coordinate axes. Express $f_{930}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{930}(12)$.



- **A** 70
- **B** 57
- **C** 65 ✓
- **D** 58

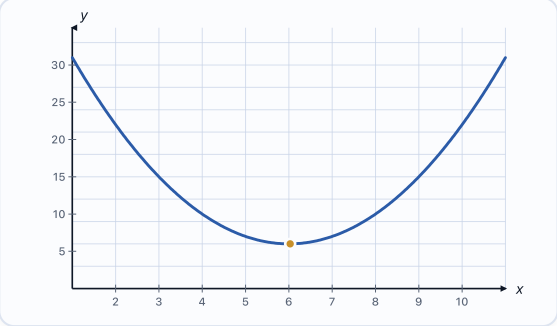
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{930}(x) = (x-5)^2 + 8$, so $k = 8$. **[M1]**
2. Evaluate $f_{930}(12) = (12-5)^2 + 8 = 49 + 8 = 57$. **[M1]**
3. Sum: $k + f_{930}(12) = 8 + 57 = 65$. **[A1]**

15 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #931), consider the function $f_{931}(x) = x^2 - 12x + 45$ illustrated on the coordinate axes. Express $f_{931}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{931}(14)$.



A 88

B 73

C 82 ✓

D 74

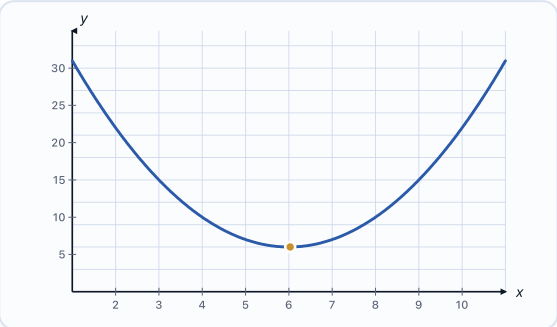
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{931}(x) = (x-6)^2 + 9$, so $k = 9$. **[M1]**
2. Evaluate $f_{931}(14) = (14-6)^2 + 9 = 64 + 9 = 73$. **[M1]**
3. Sum: $k + f_{931}(14) = 9 + 73 = 82$. **[A1]**

16 MCQ

Examining Trigonometry, Polar & Conic Sections (problem set #932), consider the function $f_{932}(x) = x^2 - 12x + 46$ illustrated on the coordinate axes. Express $f_{932}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{932}(15)$.



A 101 ✓

B 91

C 92

D 107

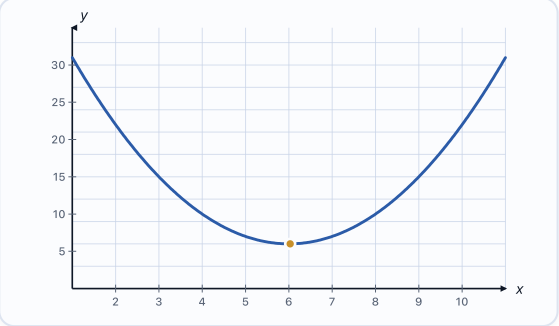
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{932}(x) = (x-6)^2 + 10$, so $k = 10$. **[M1]**
2. Evaluate $f_{932}(15) = (15-6)^2 + 10 = 81 + 10 = 91$. **[M1]**
3. Sum: $k + f_{932}(15) = 10 + 91 = 101$. **[A1]**

17 MCQ

For the configuration in Trigonometry, Polar & Conic Sections (problem set #933), consider the function $f_{933}(x) = x^2 - 12x + 47$ illustrated on the coordinate axes. Express $f_{933}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{933}(16)$.



A 128

B 111

C 122 ✓

D 112

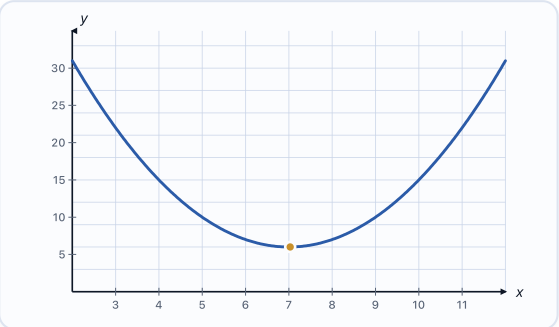
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{933}(x) = (x - 6)^2 + 11$, so $k = 11$. **[M1]**
2. Evaluate $f_{933}(16) = (16 - 6)^2 + 11 = 100 + 11 = 111$. **[M1]**
3. Sum: $k + f_{933}(16) = 11 + 111 = 122$. **[A1]**

18 MCQ

Analyzing Trigonometry, Polar & Conic Sections (problem set #934), consider the function $f_{934}(x) = x^2 - 14x + 61$ illustrated on the coordinate axes. Express $f_{934}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{934}(9)$.



A 35

B 16

C 28 ✓

D 26

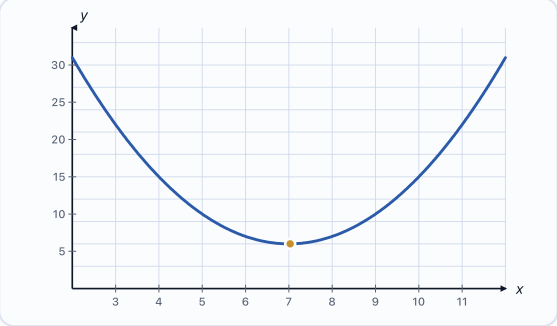
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{934}(x) = (x - 7)^2 + 12$, so $k = 12$. **[M1]**
2. Evaluate $f_{934}(9) = (9 - 7)^2 + 12 = 4 + 12 = 16$. **[M1]**
3. Sum: $k + f_{934}(9) = 12 + 16 = 28$. **[A1]**

19 MCQ

Within Trigonometry, Polar & Conic Sections (problem set #935), consider the function $f_{935}(x) = x^2 - 14x + 62$ illustrated on the coordinate axes. Express $f_{935}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{935}(10)$.



- **A** 35 ✓
- **B** 42
- **C** 22
- **D** 32

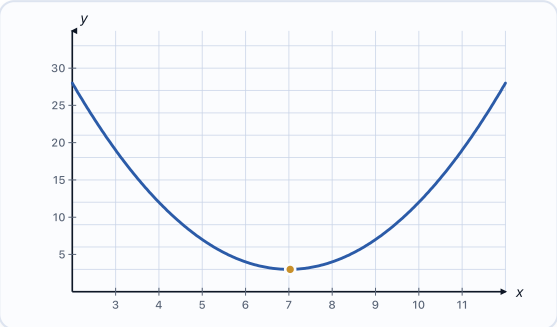
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{935}(x) = (x-7)^2 + 13$, so $k = 13$. **[M1]**
2. Evaluate $f_{935}(10) = (10-7)^2 + 13 = 9 + 13 = 22$. **[M1]**
3. Sum: $k + f_{935}(10) = 13 + 22 = 35$. **[A1]**

20 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #936), consider the function $f_{936}(x) = x^2 - 14x + 52$ illustrated on the coordinate axes. Express $f_{936}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{936}(11)$.



- **A** 22 ✓
- **B** 18
- **C** 29
- **D** 19

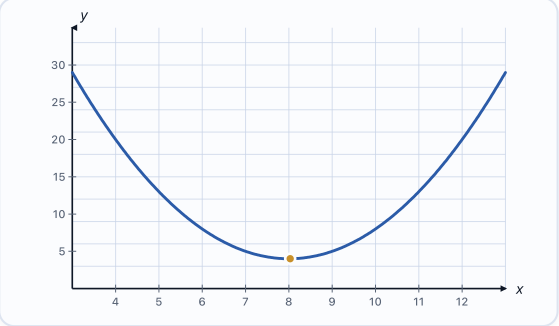
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{936}(x) = (x-7)^2 + 3$, so $k = 3$. **[M1]**
2. Evaluate $f_{936}(11) = (11-7)^2 + 3 = 16 + 3 = 19$. **[M1]**
3. Sum: $k + f_{936}(11) = 3 + 19 = 22$. **[A1]**

21 MCQ

Examining Trigonometry, Polar & Conic Sections (problem set #937), consider the function $f_{937}(x) = x^2 - 16x + 68$ illustrated on the coordinate axes. Express $f_{937}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{937}(13)$.



A 41

B 29

C 33 ✓

D 28

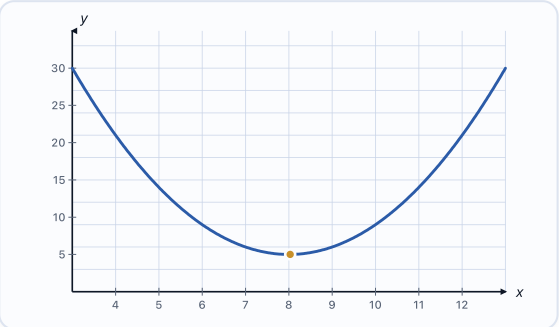
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{937}(x) = (x-8)^2 + 4$, so $k = 4$. **[M1]**
2. Evaluate $f_{937}(13) = (13-8)^2 + 4 = 25 + 4 = 29$. **[M1]**
3. Sum: $k + f_{937}(13) = 4 + 29 = 33$. **[A1]**

22 MCQ

For the configuration in Trigonometry, Polar & Conic Sections (problem set #938), consider the function $f_{938}(x) = x^2 - 16x + 69$ illustrated on the coordinate axes. Express $f_{938}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{938}(14)$.



A 40

B 46 ✓

C 54

D 41

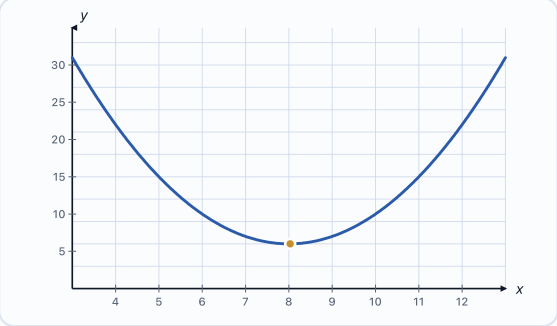
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{938}(x) = (x-8)^2 + 5$, so $k = 5$. **[M1]**
2. Evaluate $f_{938}(14) = (14-8)^2 + 5 = 36 + 5 = 41$. **[M1]**
3. Sum: $k + f_{938}(14) = 5 + 41 = 46$. **[A1]**

23 MCQ

Analyzing Trigonometry, Polar & Conic Sections (problem set #939), consider the function $f_{939}(x) = x^2 - 16x + 70$ illustrated on the coordinate axes. Express $f_{939}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{939}(15)$.



- **A** 61 ✓
- **B** 69
- **C** 55
- **D** 54

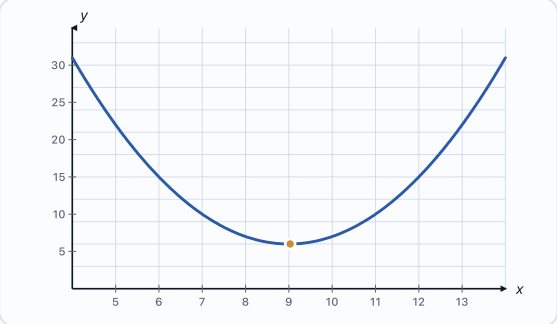
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{939}(x) = (x-8)^2 + 6$, so $k = 6$. **[M1]**
2. Evaluate $f_{939}(15) = (15-8)^2 + 6 = 49 + 6 = 55$. **[M1]**
3. Sum: $k + f_{939}(15) = 6 + 55 = 61$. **[A1]**

24 MCQ

Within Trigonometry, Polar & Conic Sections (problem set #940), consider the function $f_{940}(x) = x^2 - 18x + 88$ illustrated on the coordinate axes. Express $f_{940}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{940}(17)$.



- **A** 87
- **B** 71
- **C** 70
- **D** 78 ✓

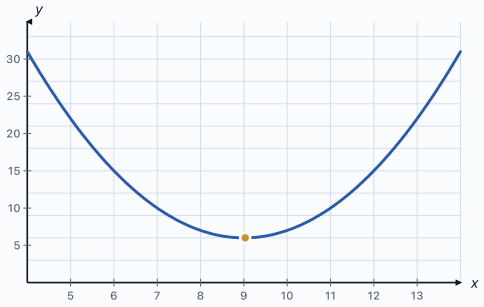
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{940}(x) = (x-9)^2 + 7$, so $k = 7$. **[M1]**
2. Evaluate $f_{940}(17) = (17-9)^2 + 7 = 64 + 7 = 71$. **[M1]**
3. Sum: $k + f_{940}(17) = 7 + 71 = 78$. **[A1]**

25 MCQ

Considering Trigonometry, Polar & Conic Sections (problem set #941), consider the function $f_{941}(x) = x^2 - 18x + 89$ illustrated on the coordinate axes. Express $f_{941}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{941}(18)$.



A 106

B 89

C 97 ✓

D 88

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Trigonometry, Polar & Conic Sections):

1. Complete the square: $f_{941}(x) = (x-9)^2 + 8$, so $k = 8$. **[M1]**
2. Evaluate $f_{941}(18) = (18-9)^2 + 8 = 81 + 8 = 89$. **[M1]**
3. Sum: $k + f_{941}(18) = 8 + 89 = 97$. **[A1]**